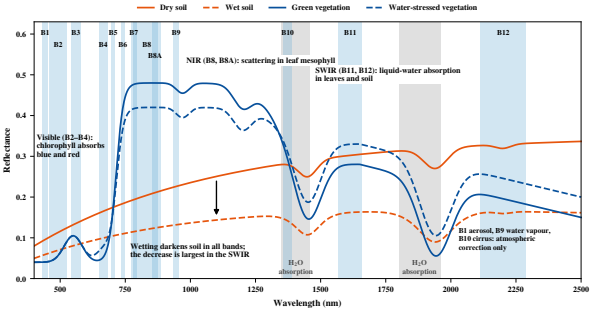
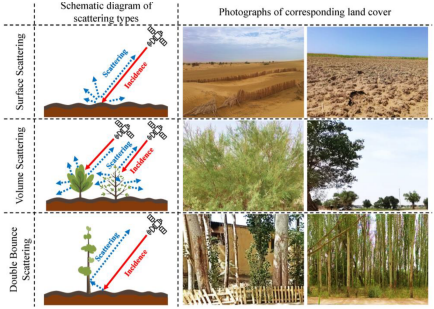
Motivation Different ways of obtaining soil moisture

Source	How SM is obtained	Depth	Main limitation
Microwave (SMAP, SMOS, ASCAT, S1)	invert: σ^0 or $T_B = f(\text{SM}, \text{roughness}, \text{vegetation})$	top $\sim$5 cm; informs part of the root zone (Feldman et al., 2023)	shallow; assumes vegetation and roughness; coarse (9–40 km) except S1
Optical / thermal (S2, Landsat, MODIS)	empirical relations with surface properties: reflectance and indices; LST–vegetation space (e.g. triangle method, TVDI)	skin / surface	indirect, site-specific calibration; clouds; canopy hides the soil
Land-surface models	$d\theta/dt = P - ET - Q - D$, with parameters p	full profile	forcing errors, soil and vegetation parameters, process representation (Koster et al., 2009; Seneviratne et al., 2010)
Reanalysis (ERA5-Land)	LSM driven by reanalysis meteorology	full profile	model assumptions dominate deep layers with few observational constraints (Li et al., 2021; Miguez-Macho and Fan, 2021)
In situ (ISMN)	direct probe measurement	several depths	point scale; sparse, uneven global coverage
Data-driven (ML / DL)	SM $= f_\theta(\text{inputs})$, learned from in-situ labels; no ad-hoc process assumptions	any depth with labels	needs enough, representative sites; generalisation?; hard to interpret?

Motivation Multimodal remote sensing



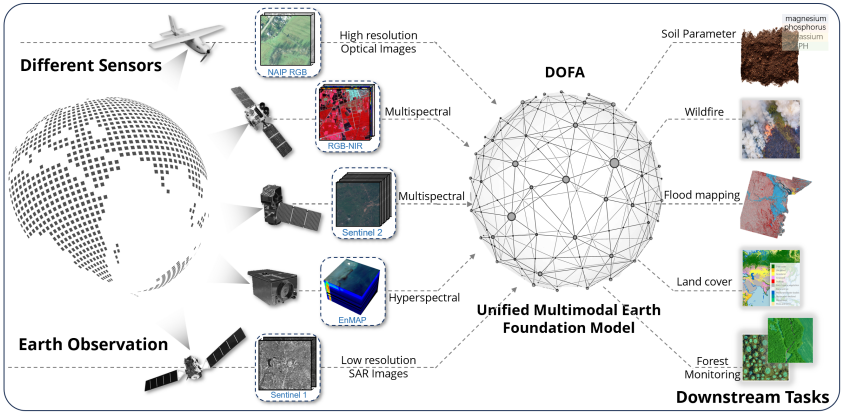
Optical: dry vs wet soil; green vs water-stressed vegetation



SAR scattering types (Nurmemet et al., 2025)

Soil and vegetation water change both optical reflectance and SAR backscatter

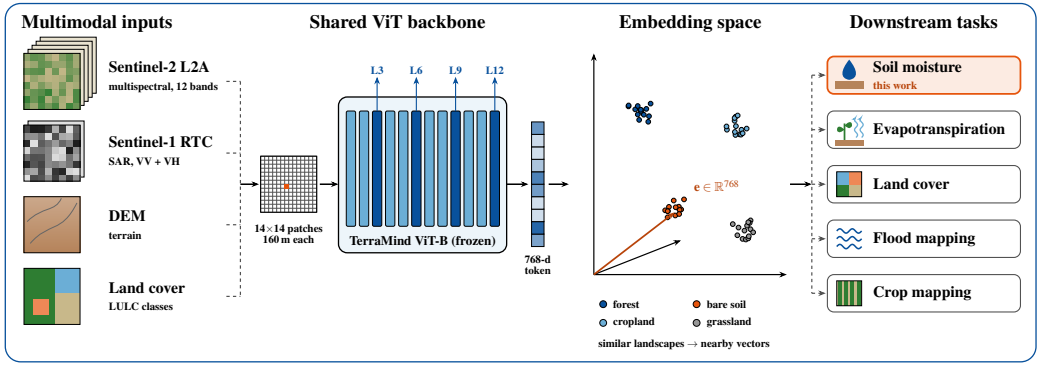
Motivation Geospatial foundation models (GFMs)



DOFA: one model for many sensors and many downstream tasks (Xiong et al., 2024, arXiv:2403.15356)

Once pre-trained on multimodal, unlabelled data, it works for many downstream tasks

Motivation TerraMind embeddings for soil moisture

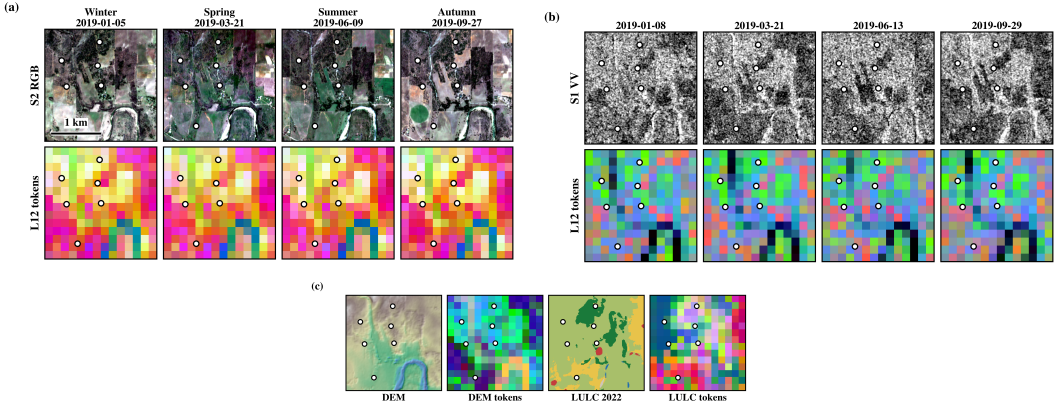


S2, S1, DEM and LULC → frozen TerraMind ViT-B → one 768-d embedding per 160 m patch → downstream tasks

The same embedding can serve many tasks; in this case, soil moisture

Motivation

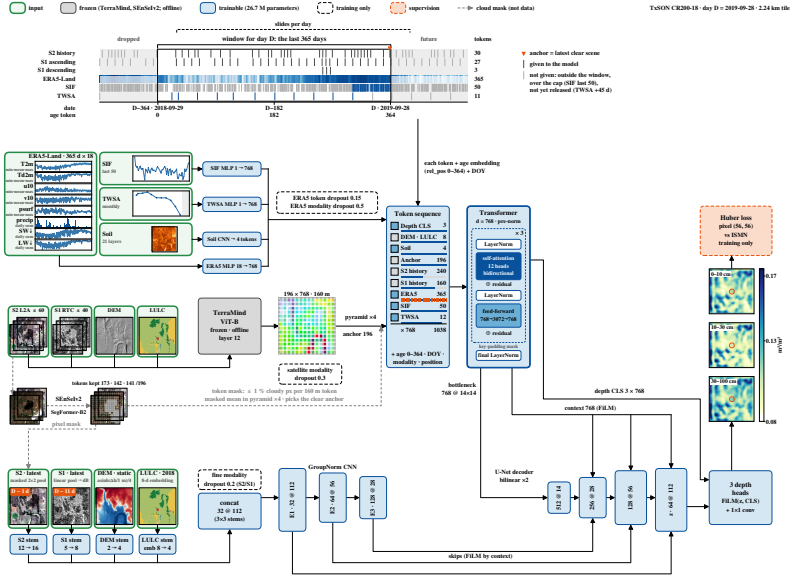
Example: TerraMind embeddings at one site (TxSON, 2019)



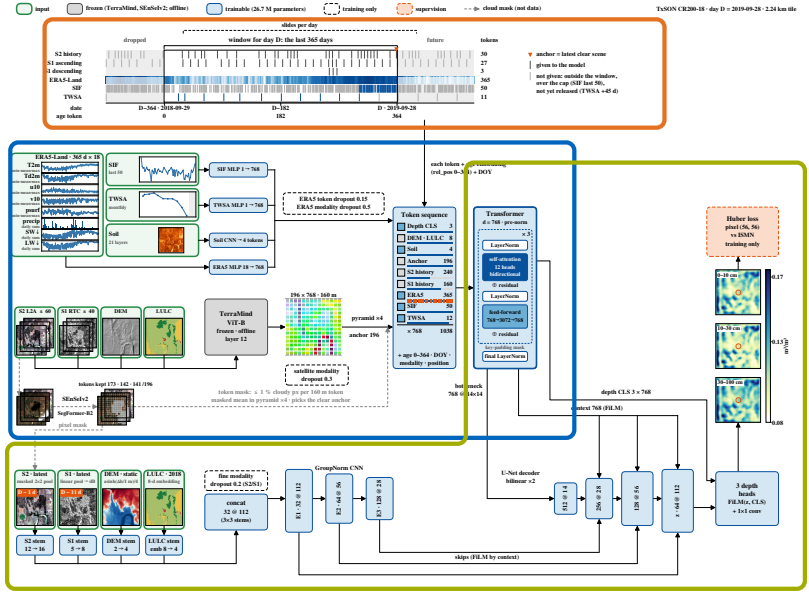
(a) S2 in four seasons, (b) S1 VV, (c) DEM and LULC, each with its TerraMind L12 tokens

Each 160 m patch → one 768-d vector; colours show its first 3 principal components

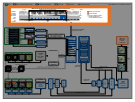
Method Model architecture

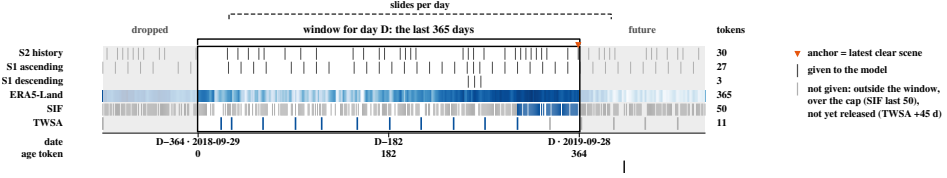


Method Model architecture

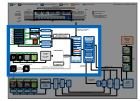


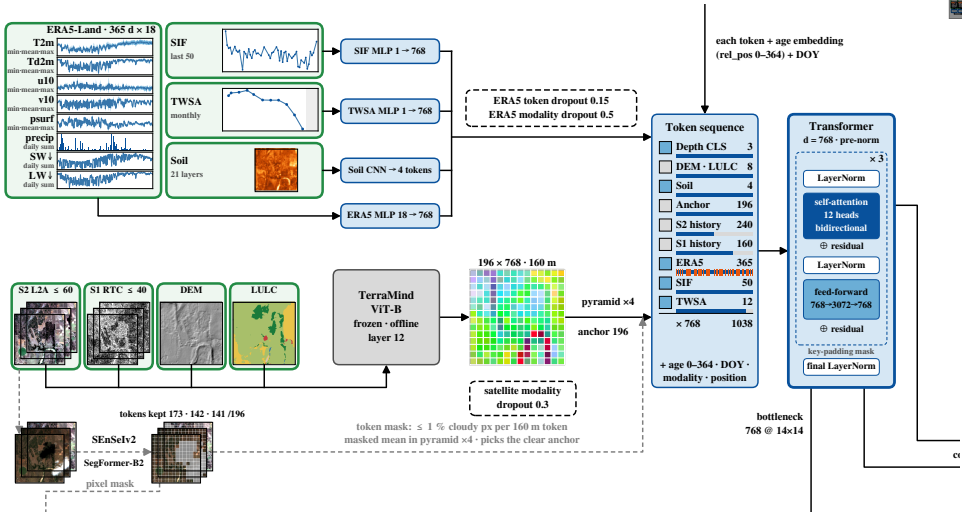
Method Inputs: what the model sees for day D



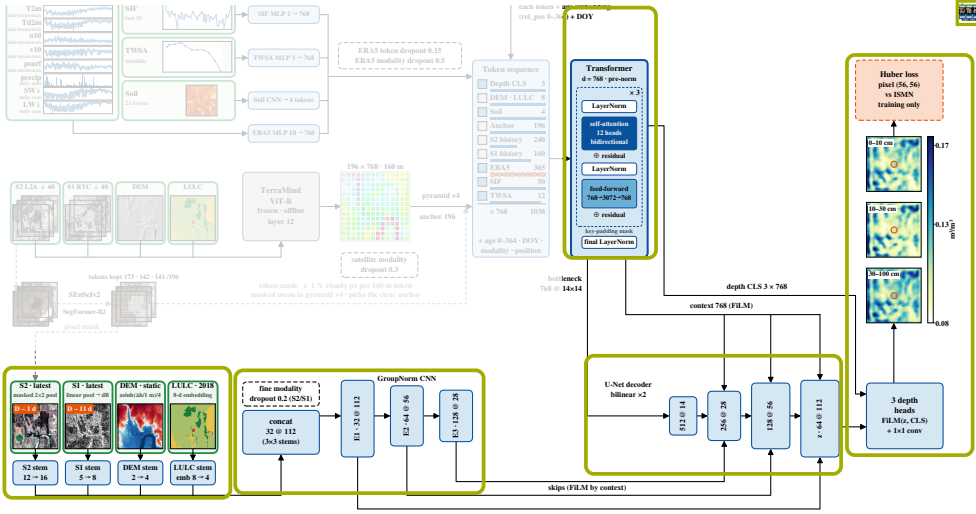


Method Coarse path: tokens → transformer

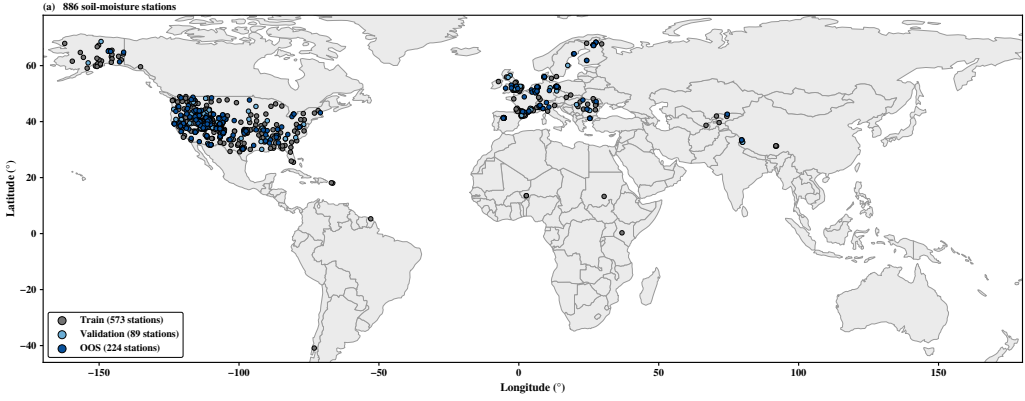




Method **Fine path: 20 m CNN → U-Net decoder → depth heads**



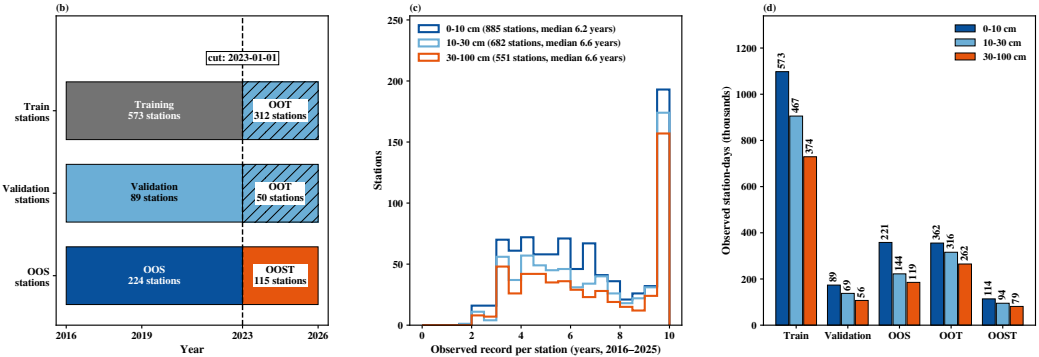
Data In-situ soil moisture: global coverage



886 soil-moisture stations (ISMN and flux sites): train 573, validation 89, out-of-sample (OOS) 224

Dense in North America and Europe; sparse in the tropics and the Southern Hemisphere

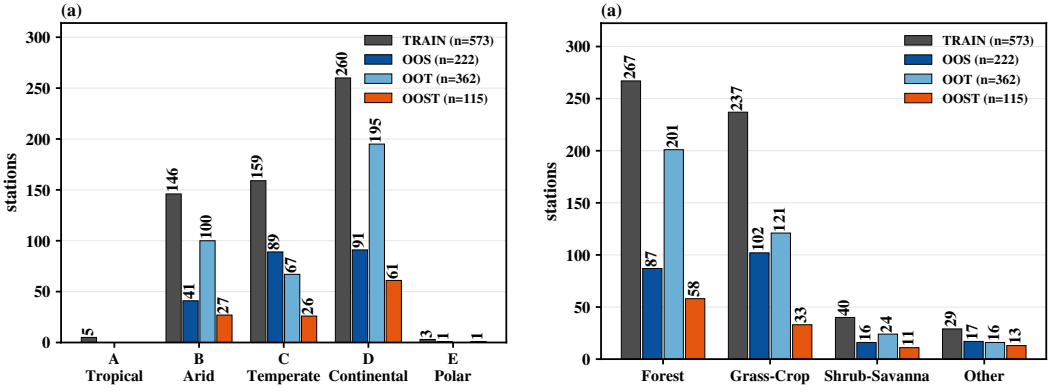
Data Depth coverage, record length and splits



(b) train / validation / OOS stations, time cut 2023-01-01; (c) observed record per station; (d) observed station-days

0–10 cm: 885 stations (median 6.2 yr); 10–30 cm: 682 (6.6 yr); 30–100 cm: 551 (6.6 yr)

Data Climate and land-cover coverage



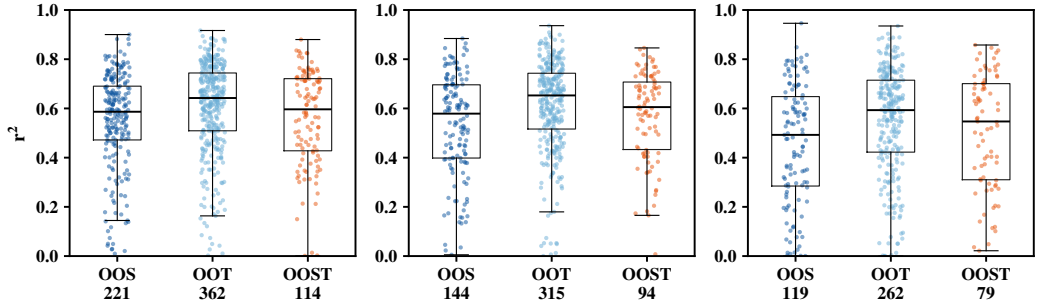
Stations per Köppen climate zone (left) and land-cover class (right), by split

Mostly continental, temperate and arid; forest and grass–crop; almost no tropics or polar

Stations	990 in inventory; **886** with soil moisture (train 573 / val 89 / OOS 224)
Depths	0–10 cm: 885 10–30 cm: 682 30–100 cm: 551 stations
Period	2016–2025; out-of-time test from 2023-01-01
Soil-moisture observations	**5.14 M** observed daily values (2.10 M / 1.68 M / 1.37 M per depth)
Sentinel-2 L2A	272,281 scenes (994 sites); cloud $< 75\,\%$ per scene, cloudy tokens masked
Sentinel-1 RTC	496,835 scenes: 251,310 ascending + 245,525 descending
ERA5-Land	18 variables, 2.46 M station-days (990 stations)
Static / other	DEM, LULC, soil (OpenLandMap), SIF, TWSA

About 5.1 million daily soil-moisture observations at 886 stations, 0–100 cm

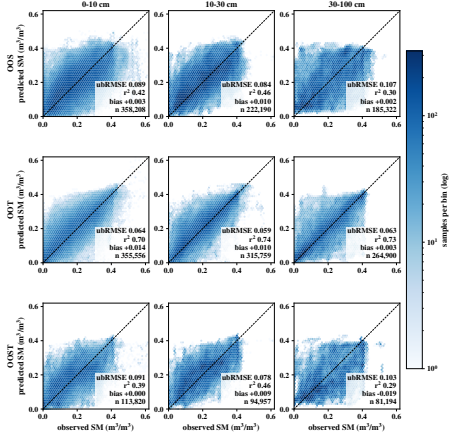
Results Per-station r^2 by depth



Columns: 0–10, 10–30, 30–100 cm; one point per station; OOS, OOT (2023–2025), OOST (2023–2025)

Median r^2 about 0.6 near the surface, lower at 30–100 cm for new stations (0.49)

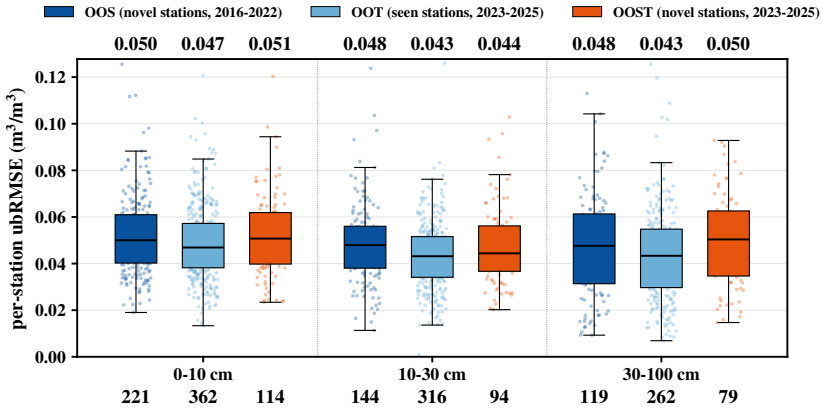
Results Predicted vs observed, by split and depth



All daily samples; rows OOS (new stations), OOT (known stations, 2023–2025), OOST (new stations, 2023–2025)

New years at known sites (OOT, r^2 0.70–0.74) fit far better than new sites (r^2 0.29–0.46)

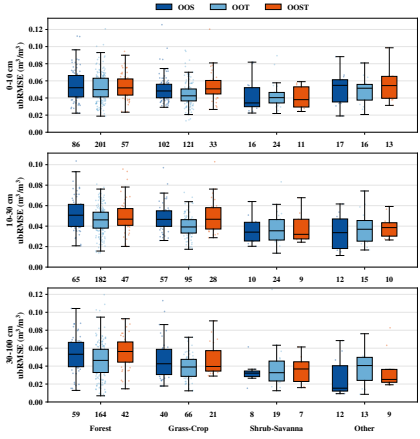
Results Per-station ubRMSE by depth



One point per station; medians above, station counts below

Median per-station ubRMSE 0.043–0.051 m^3/m^3 at every depth and split

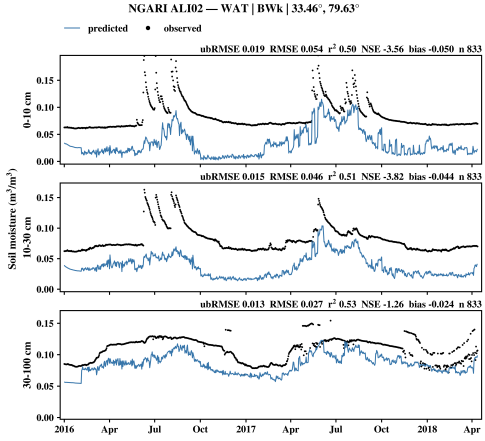
Results Per-station ubRMSE by land cover



Rows: depths; groups: land-cover class; colours: OOS, OOT, OOST

Similar error across land cover; forest slightly highest, shrub–savanna lowest

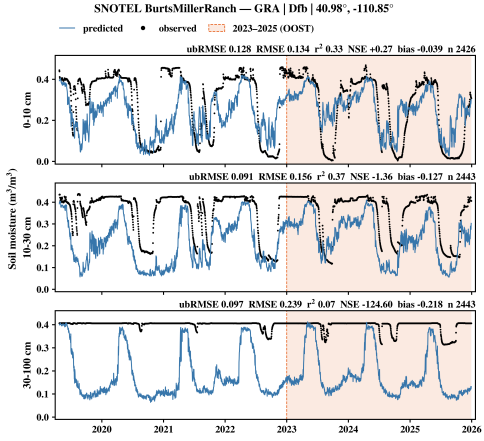
Results Best site: NGARI ALI02 (Tibetan Plateau)



Lowest OOS ubRMSE among stations observed at all three depths; blue predicted, black observed

Dynamics right at all depths (ubRMSE 0.019 / 0.015 / 0.013), but biased low by 0.02–0.05

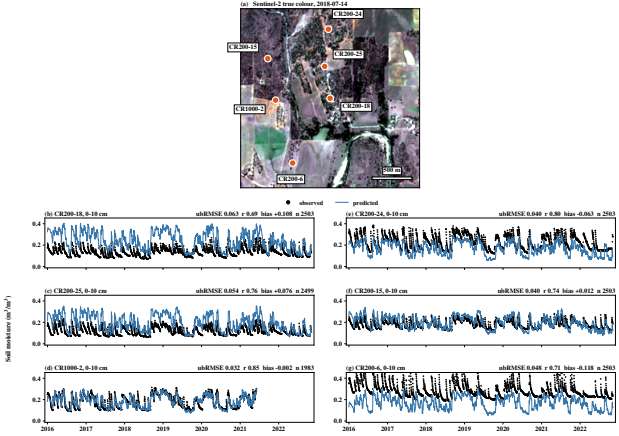
Results Worst site: SNOTEL Burts Miller Ranch (Utah)



Highest OOS ubRMSE among stations observed at all three depths; shaded = 2023–2025 (OOST)

Snow-fed site: the deep layer stays near saturation, but the model dries it every summer

Results TxSON: six stations inside one 2.2 km tile



(a) Sentinel-2 true colour with the six stations; (b–g) 0–10 cm time series, predicted vs observed

Timing right at every station (r 0.69–0.85); offsets remain (bias −0.12 to +0.11)